Luxury Housing Sales Analysis – Bengaluru

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1. Introduction

The real estate sector is a crucial part of urban economic growth. This project focuses on analyzing luxury housing sales data in Bengaluru to uncover trends, patterns, and insights that can help businesses make informed decisions.

The project demonstrates an end-to-end data analytics pipeline using **Python, SQL, and Power BI**, simulating a real-world business intelligence workflow.

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2. Problem Statement

To build a complete analytics solution that:

- Cleans and processes raw housing data
- Stores structured data in a SQL database
- Creates an interactive dashboard using Power BI
- Generates meaningful business insights

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3. Dataset Description

- Dataset Name: Luxury Housing Bangalore
- Size: 100,000+ rows
- Key Features:

Column	Description
Property_ID	Unique property identifier
Micro_Market	Area/location in Bengaluru

Developer_Name	Builder name
Configuration	Type (2BHK, 3BHK, etc.)
Ticket_Price_Cr	Property price
Amenity_Score	Facilities rating
Transaction_Type	Booking status
Purchase_Quarter	Time period
Sales_Channel	Sales source
Buyer_Type	Customer category

Data Cleaning & Preprocessing (Python)

Steps performed:

- Removed duplicates
- Converted string columns to numeric
- Handled missing values using median
- Standardized text fields
- Created new features:

Price_per_Sqft
Booking_Flag
Quarter_Number

Key Fixes:

- Converted `Ticket_Price_Cr` from string → numeric
- Fixed incorrect Booking_Flag logic
- Extracted Quarter_Number correctly

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5. SQL Integration

Database Created:

luxury_housing_db

Table:

luxury_housing

Key Queries:

- Total Records:

```
SELECT COUNT(*) FROM luxury_housing;
```

- Total Revenue:

```
SELECT SUM(Ticket_Price_Cr) FROM luxury_housing;
```

- Booking Distribution:

```
SELECT Booking_Flag, COUNT(*) FROM luxury_housing GROUP BY Booking_Flag;
```

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6. Power BI Dashboard

An interactive dashboard was built with:

KPI Cards:

- Total Revenue
- Average Price
- Total Properties
- Booking Conversion Rate

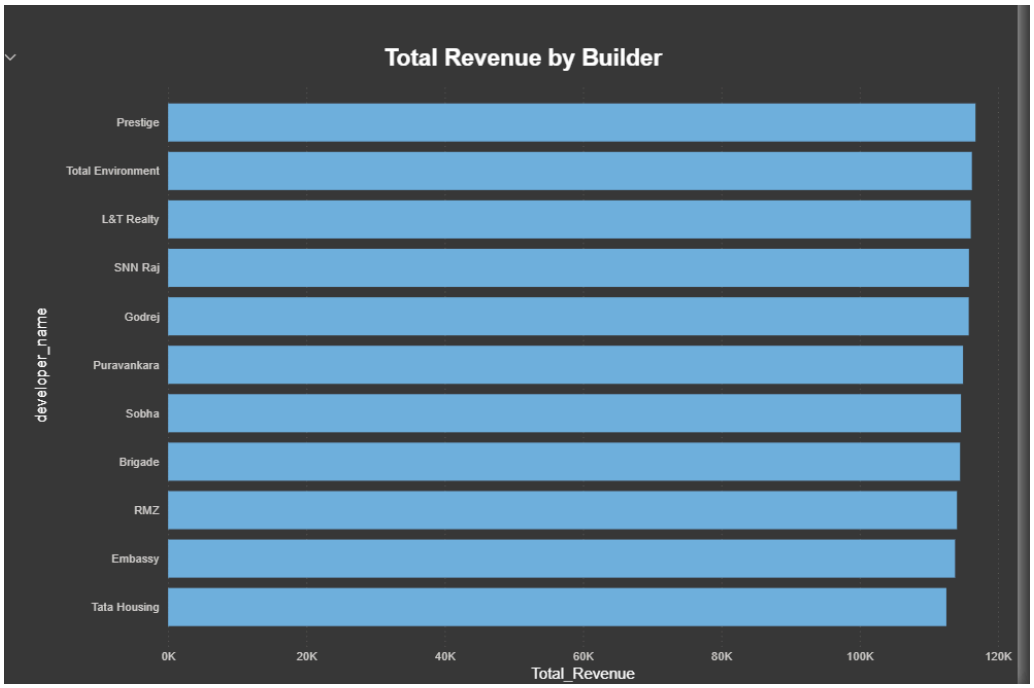
Visuals:

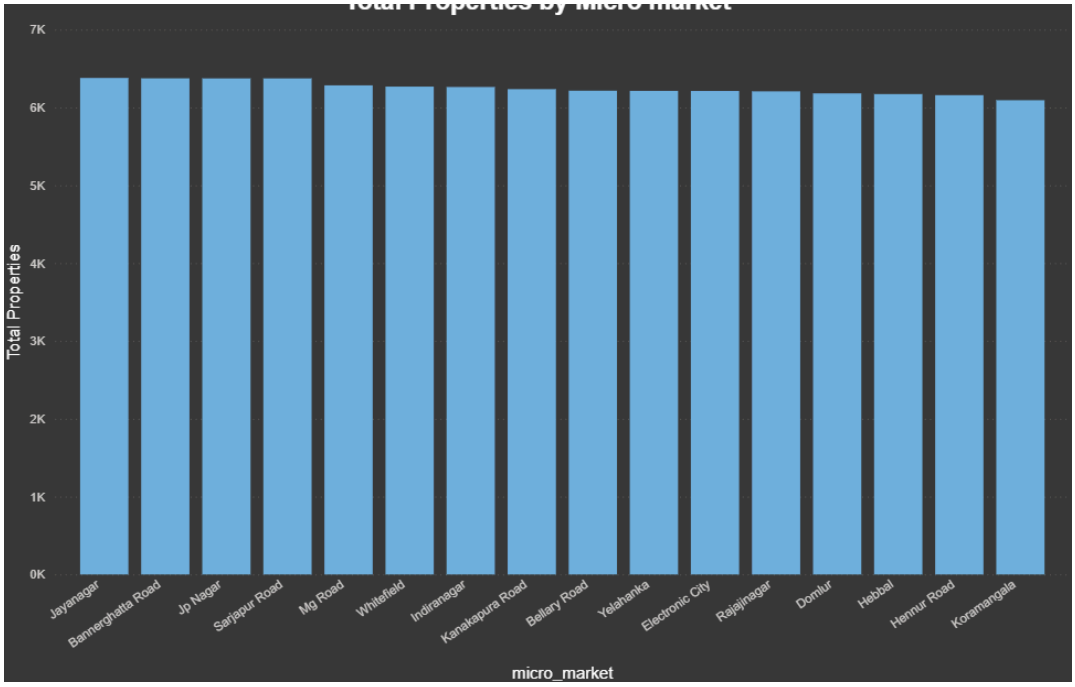
- Revenue by Builder
- Quarterly Booking Trend
- Configuration Demand
- Amenity vs Booking

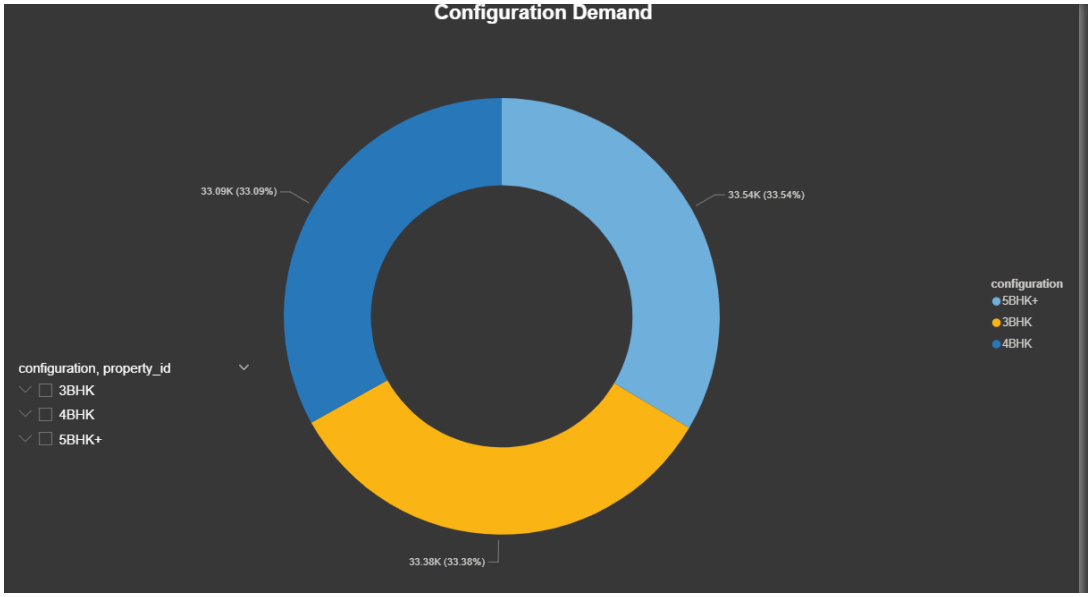
- Sales Channel Performance
- Micro-Market Distribution
- Geographical Map

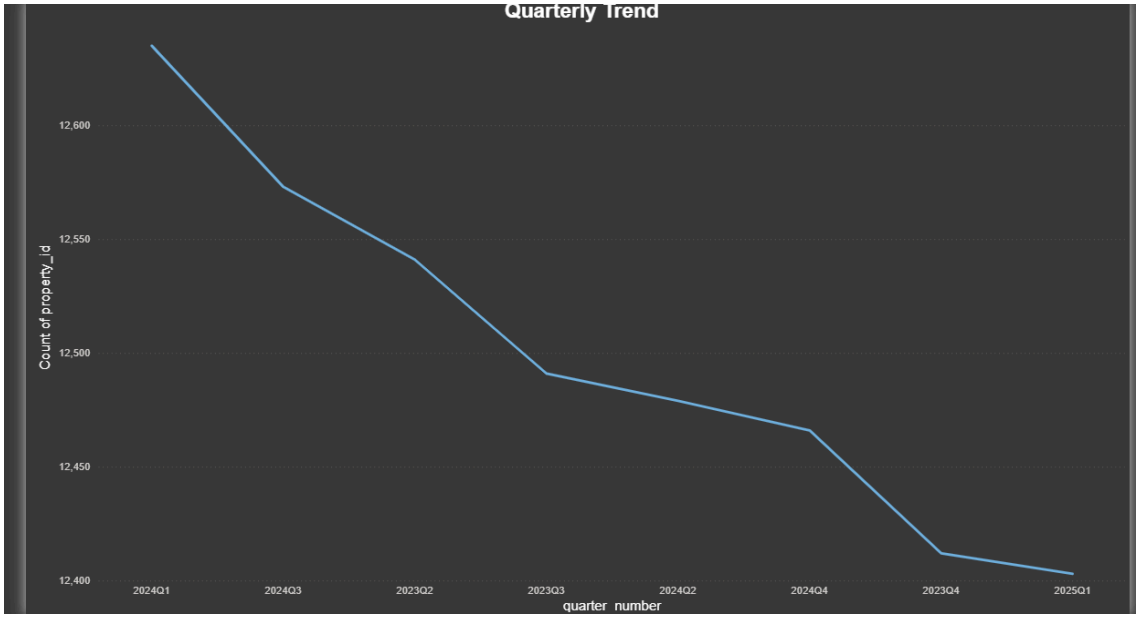
Filters (Slicers):

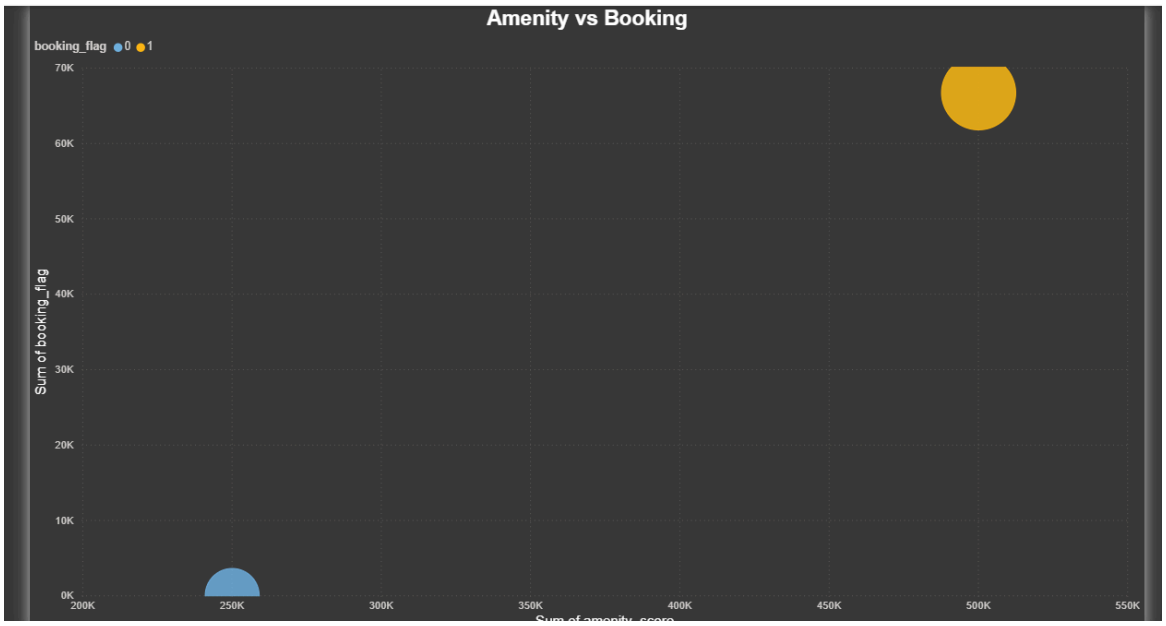
- Developer
- Micro Market
- Quarter
- Configuration
- Buyer Type

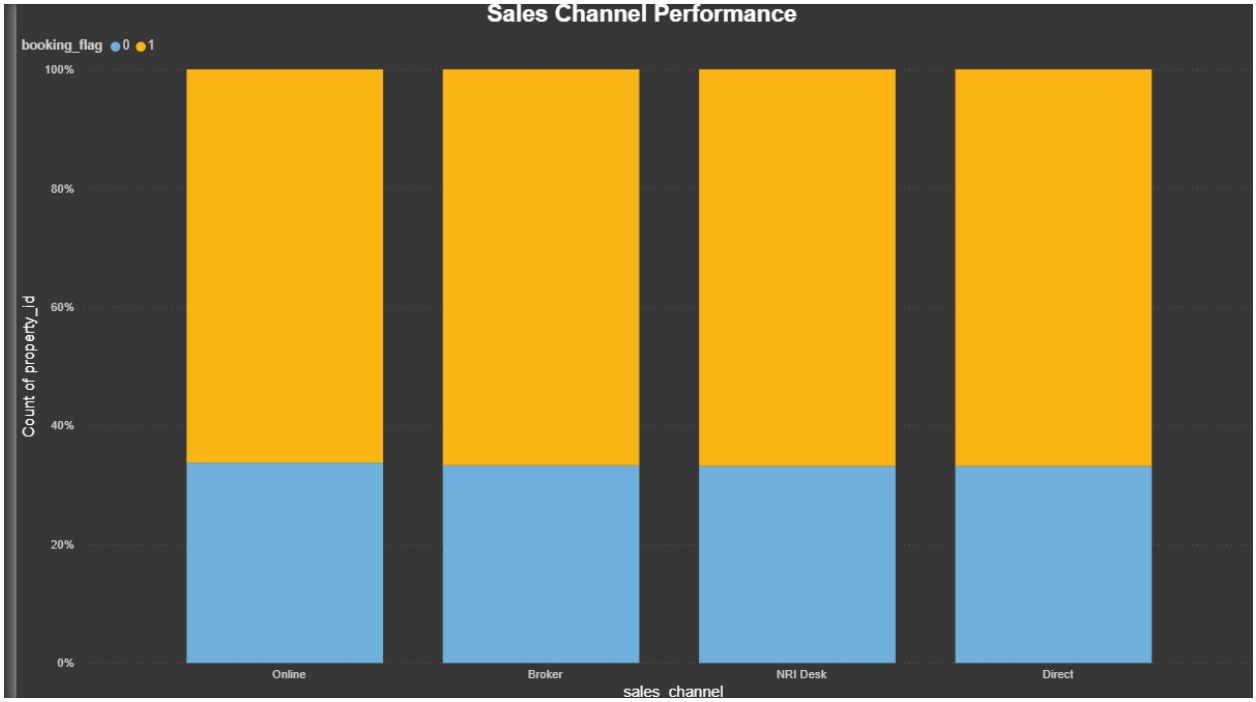


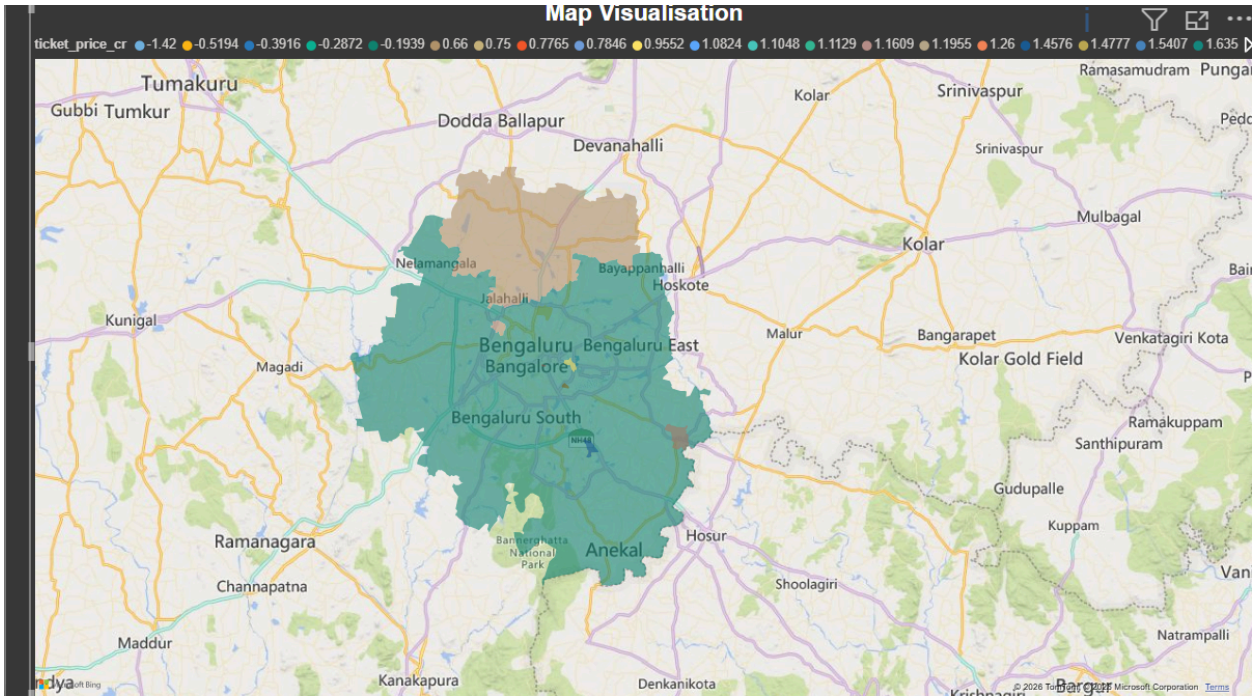

Total Properties by Micro Market
7K
6K
5K
4K
3K
2K
1K
0K
Total Properties
Jayanagar
Bannerghatta Road
Jp Nagar
Sarjapur Road
Mg Road
Whitefield
Indiranagar
Kanakapura Road
Bellary Road
Yelahanka
Electronic City
Rajajinagar
Domlur
Hebbal
Hennur Road
Koramangala
micro_market


Configuration Demand
33.09K (33.09%)
33.54K (33.54%)
33.38K (33.38%)
configuration, property_id
3BHK
4BHK
5BHK+
configuration
5BHK+
3BHK
4BHK


Quarterly Trend
Count of property_id
12,600
12,550
12,500
12,450
12,400
2024Q1
2024Q3
2023Q2
2023Q3
2024Q2
2024Q4
2023Q4
2025Q1
quarter_number


Amenity vs Booking
booking_flag 0 1
Sum of booking_flag
70K
60K
50K
40K
30K
20K
10K
0K
200K
250K
300K
350K
400K
450K
500K
550K


Sales Channel Performance
booking_flag 0 1
100%
80%
60%
40%
20%
0%
Count of property_id
Online
Broker
NRI Desk
Direct
sales_channel


Map Visualisation
ticket_price_cr -1.42 -0.5194 -0.3916 -0.2872 -0.1939 0.66 0.75 0.7765 0.7846 0.9552 1.0824 1.1048 1.1129 1.1609 1.1955 1.26 1.4576 1.4777 1.5407 1.635
Tumakuru
Gubbi Tumkur
Dodda Ballapur
Devanahalli
Kolar
Srinivaspur
Ramasamudram
Mulbagal
Nelamangala
Bayappanhalli
Hoskote
Jalahalli
Kunigal
Magadi
Bengaluru
Bangalore
Bengaluru East
Malur
Bangarapet
Kolar Gold Field
Venkatagiri Kota
Bengaluru South
Ramakuppam
Santhipuram
Gudupalle
Kuppam
Ramanagara
Anekal
Hosur
Channapatna
Shoolagiri
Maddur
Natrampalli
Kanakapura
Denkanikota

7. Exploratory Data Analysis (EDA)

Performed analysis to understand:

- Data distribution
- Missing values
- Feature relationships
- Market segmentation

EDA helped in identifying patterns before visualization.

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8. Business Insights

- Top developers contribute the majority of total revenue
- 3BHK and 4BHK configurations dominate market demand
- Higher amenity scores lead to better booking conversion
- Certain micro-markets consistently outperform others
- Booking trends vary across quarters indicating seasonal demand
- Sales channel effectiveness differs significantly

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9. Storytelling

The dashboard tells a clear business story:

- Market is driven by key developers
- Demand is concentrated in specific configurations
- Location plays a major role in performance
- Amenities influence customer decisions
- Seasonal trends impact booking patterns

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10. Real Estate Market Analytics

The project provides insights into:

- Pricing strategies
- Demand patterns
- Location-based performance
- Builder competitiveness
- Customer behavior

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11. Conclusion

This project successfully demonstrates a real-world analytics pipeline by integrating:

- Python for data cleaning & Engineering
- SQL for data management
- Power BI for visualization

The insights generated can help real estate companies optimize pricing, marketing strategies, and investment decisions.

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12. Tools & Technologies

- Python (Pandas, NumPy)
- MySQL
- Power BI
- SQLAlchemy

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13. Deliverables

- Cleaned dataset
- Python script
- SQL queries
- Power BI dashboard
- Project documentation